Type Certificate

Certificate

TC-201206 Rev.1

Issue date

2025-03-21

Expiration date

2025-12-25

DAkkS
Deutsche
Akkreditierungsstelle
D-ZE-11326-01-00

Approved for issue on
behalf of DEWI-OCC GmbH

Federico Osvald
Vice Head of Certification Body
Bremen, 2025-03-21

This certificate is issued to

ENERCON GmbH

Dreekamp 5
Aurich, 26605
Germany

for the wind turbine

E-82 E2

This certificate attests compliance with the wind turbine class and main standard

IIA, IEC 61400-1:2005+AMD1:2010

The conformity evaluation was carried out in accordance with the rules and procedures of

IEC 61400-22 Ed.1.0

The certificate is based on the reference documents listed in the subsequent pages.

The main wind turbine type characteristics are specified in the annex of this certificate.

This certificate is valid subject to the prescribed maintenance.

Changes in the system design or the manufacturer's quality system are to be approved by the Certification Body. Without approval, the certificate loses its validity.

This Certificate is based on the following reference documents:

Design evaluation conformity statement
DEWI-OCC GmbH: Conformity Statement
“Design Evaluation Conformity Statement - Wind Turbine E-82 E2”,
Doc. No. STC-150607 Rev. 3, 2019-05-10

Manufacturing evaluation conformity statement
DEWI-OCC GmbH: Conformity Statement
“Manufacturing Conformity Statement - Wind Turbine E-82 E2”,
Doc. No. STC-150608 Rev. 5, 2023-07-13

Type test evaluation conformity statement
DEWI-OCC GmbH: Conformity Statement
“Type Test Conformity Statement - Wind Turbine E-82 E2”,
Doc. No. STC-150609 Rev. 2, 2019-05-10

Final evaluation report
DEWI-OCC GmbH: Report
“Final Evaluation Report - Wind Turbine E-82 E2”,
Doc. No. R101332-12 Rev. 5, 2025-03-21

DEWI-OCC Offshore and Certification Centre GmbH
Contrescarpe 45
28195 Bremen, Germany

Annex - wind turbine type characteristics

General	
Customer	ENERCON GmbH Dreekamp 5 Aurich, 26605 Germany
Designer	ENERCON GmbH Dreekamp 5 Aurich, 26605 Germany
WT class	II A
Machine parameters:	
Power Regulation:	Pitch control
Rotor Orientation:	Upwind
Number of rotor blades:	3
Rotor Tilt:	5°
Cone Angle:	0°
Rated power:	2000 kW / 2300 kW
Rotor diameter:	82 m
Hub height(s):	78 m, 84 m, 98 m, 108 m, 138 m
Design life time:	20 years
Wind conditions:	
Cut-in wind speed	3 m/s
Rated wind speed	13.4 m/s
Range of reduced power (storm control)	24 m/s – 34 m/s
Cut-out wind speed (10 min mean)	34 m/s
Characteristic turbulence intensity I_{ref} at V_{hub} = 15m/s:	16%
Annual average wind speed at hub height V_{ave}:	8.5 m/s
Reference wind speed V_{ref}:	42.5 m/s
Hub height 50-year extreme wind speed Ve50:	59.5 m/s
Electrical network conditions:	
Normal supply voltage:	10 kV – 30 kV
Normal supply frequency:	50/60 Hz

Major components: ** If not otherwise stated, the certificate holder is the manufacturer		**Rotor Blade Type E82-2**	**Rotor Blade Type E-82 EP2-RB-05**
Blade:			
Type: Material: Blade length: Manufacturer:	E82-2 Fibre reinforced plastics 38.8 m ENERCON GmbH	√ √ √ √	
Type: Material: Blade length: Manufacturer:	E 82-2 SEP (Trailing Edge Serrations) Fibre reinforced plastics 38.8 m ENERCON GmbH	√ √ √ √	
Type: Material: Blade length: Manufacturer:	E-82 EP2-RB-05 Glass fiber reinforced epoxy resin 38.8 m ENERCON GmbH		√ √ √ √
Pitch gearbox:			
Supplier: Type: Ratio:	Liebherr Three independent pitch system 131.3	√ √ √	√ √ √
Supplier: Type: Ratio:	Zollern Three independent pitch system 131.3	√ √ √	
Supplier: Type: Ratio:	Brevini Power Transmission Three independent pitch system 130.1	√ √ √	
Supplier: Type: Ratio:	ABM - Greiffenberger Antriebstechnik GmbH Three independent pitch system 131.3	√ √ √	
Pitch bearings:			
Supplier: Type:	Liebherr Ball Bearing Slewing Ring	√ √	√ √
Supplier: Type:	PSL Ball Bearing Slewing Ring	√ √	
Supplier: Type:	Rothe Erde Ball Bearing Slewing Ring	√ √	

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		Rotor Blade Type E82-2	Rotor Blade Type E-82 EP2-RB-05
Supplier: Type:	Hanse Drehverbindung Ball Bearing Slewing Ring	√ √	√ √
Supplier: Type:	La Leonessa Ball Bearing Slewing Ring	√ √	
Hub:			
Drawing-No.: Material:	82.01.054-6 EN-GJS-400-18U-LT	√ √	√ √
Drawing-No.: Material:	EP2.01.121-0 EN-GJS-400-18-LT	√ √	√ √
Main bearings:			
Supplier: Type: Drawing-No.:	PSL Double Row Tapered Roller Bearing PSL612-319-4-PV	√ √ √	√ √ √
Supplier: Type: Drawing-No.:	SKF Double Row Tapered Roller Bearing BT2-8168 HA1VK 443	√ √ √	
Supplier: Type: Drawing-No.:	TIMKEN Double Row Tapered Roller Bearing E-50931	√ √ √	
Supplier: Type: Drawing-No.:	FAG Double Row Tapered Roller Bearing F-809483.TR2 A01	√ √ √	
Supplier: Type: Drawing-No.:	NTN SNR Double Row Tapered Roller Bearing 07-17718-C, Rev.A	√ √ √	
Supplier: Type: Drawing-No.:	KOYO Double Row Tapered Roller Bearing DMA136844	√ √ √	
Supplier: Type: Drawing-No.:	PSL Single Row Cylindrical Roller Bearing 412-303-4-PV	√ √ √	√ √ √
Supplier: Type: Drawing-No.:	SKF Single Row Cylindrical Roller Bearing BC1-8033/HB1VK443	√ √ √	

		Rotor Blade Type E82-2	Rotor Blade Type E-82 EP2-RB-05
Supplier: Type: Drawing-No.:	TIMKEN Single Row Cylindrical Roller Bearing C103720 Rev. C	√ √ √	
Supplier: Type: Drawing-No.:	FAG Single Row Cylindrical Roller Bearing F-804522.01.ZL	√ √ √	
Supplier: Type: Drawing-No.:	NTN SNR Single Row Cylindrical Roller Bearing 07-17719-C, Rev. B	√ √ √	
Supplier: Type: Drawing-No.:	KOYO Single Row Cylindrical Roller Bearing SCA218980	√ √ √	
Main frame:			
Drawing-No.: Material:	82.03.115-9 EN-GJS-400-18U-LT	√ √	√ √
Rotor brake:			
Supplier: Type:	KTR EMB 300 RT	√ √	√ √
Supplier: Type:	Hanning & Kahl HEAW 300 T	√ √	√ √
Rotor lock:			
Drawing-No.: Drawing-No.:	82.00.010-0 82.02.201-2	√	 √
Yaw gearbox:			
Supplier: Type:	Liebherr DAT 400/2449-4000	√ √	
Supplier: Type:	Liebherr DAT 400/2450-4000	√ √	√ √
Supplier: Type:	Zollern ZHP 3.25-L-STZ-P	√ √	
Supplier: Type:	Brevini Power Transmission VDL2604DCL	√ √	
Supplier: Type:	Nabtesco RGS35CW	√ √	

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		Rotor Blade Type E82-2	Rotor Blade Type E-82 EP2-RB-05
Yaw bearing:			
Supplier: Drawing-No.:	Rothe Erde 091.45.2165.001.44.1502	√ √	
Supplier: Drawing-No.:	Liebherr KUD588VA801-900	√ √	√ √
Supplier: Drawing-No.:	Hanse Drehverbindungen 21452165001	√ √	
Supplier: Drawing-No.:	PSL 9E-2B45-2165-1468, Rev.5	√ √	
Supplier: Drawing-No.:	La Leonessa CB-E-2493-D-3-00	√ √	
Tower top flange:			
Drawing-No.: Material:	S 04.0002-7 S355NL	√ √	√ √
Tower:			
Hub Height: Designer: Type: Specification:	78 m ENERCON GmbH Tubular steel E-82 E2/S/77/4F/02	√ √ √ √	√ √ √ √
Hub Height: Designer: Type: Specification:	78 m ENERCON GmbH Tubular steel E-82 E2/S/77/5K/01	√ √ √ √	
Hub Height: Designer: Type: Specification:	84 m ENERCON GmbH Tubular steel E-82 E2/S/83/5K/01	√ √ √ √	
Hub Height: Designer: Type: Specification:	84 m ENERCON GmbH Hybrid E-82 E2/BF/82/17/05	√ √ √ √	
Hub Height: Designer: Type:	84 m ENERCON GmbH Precast concrete	√ √ √	

		Rotor Blade Type E82-2	Rotor Blade Type E-82 EP2-RB-05
Specification:	E-82 E2/BF/83/17/01	√	
Hub Height: Designer: Type: Specification:	84 m ENERCON GmbH Precast concrete E-82 E2/BF/83/17/02	√ √ √ √	
Hub Height: Designer: Type: Specification:	98 m ENERCON GmbH Precast concrete E-82 E2/BF/97/20/01	√ √ √ √	
Hub Height: Designer: Type: Specification:	98 m ENERCON GmbH Precast concrete E-82 E2/BF/97/20/05	√ √ √ √	
Hub Height:	108 m	√	
Designer:	ENERCON GmbH	√	
Type:	Precast concrete	√	
Specification: Hub Height: Designer: Type:	E-82 E2/BF/107/23/01 108 m ENERCON GmbH Precast concrete	√ √ √ √	
Specification:	E-82 E2/BF/107/23/05	√	
Hub Height: Designer: Type: Specification:	138 m ENERCON GmbH Precast concrete E-82 E2/BF/137/24/01	√ √ √ √	
Generator:			
Supplier: Type:	Induction Generatorfertigung GmbH Synchronous generator	√ √	√ √
Inverter:			
Type: Amount of Units	Full load inverter 7 to 9	√ √	√ √
Lightning protection system:			
Lightning protection level	LPL I	√	√

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